V. Numerical Validation

A. Benchmark, implementation and protocol

The benchmark is the class-(H) telegraph reaction–diffusion–advection lattice

$$\begin{aligned}\dot{q} &= p,\\ \dot{p} &= -(\nu L_h + \alpha I)q - \gamma p - c\,W_f D_h W_f\, q\\ &\quad - \kappa(q - \sin q) + Bu + B_w w,\end{aligned}$$

on $N_m = 160$ nodes ($N = 320$ states) with Dirichlet data, $\nu = 0.05$, $\alpha = 1$, $\gamma = 1$, $\kappa = 0.05$, two actuators ($m = 2$) and one disturbance channel. Here D_h is the skew central difference and the advection filter is

$$W_f = \Phi_f\,\mathrm{diag}(\underbrace{1,\dots,1}_{2},\, w,\dots,w)\,\Phi_f^\top,$$

where Φ_f holds the first $K_f = 4$ mass-normalised sine modes. The plain projector $\Pi_f = \Phi_f\Phi_f^\top$ is the special case $w = 1$. The cutoff-mode weight w is a disclosed benchmark parameter: it is a configuration literal, it is swept in Experiment 9, and the deployed value $w^\star = 2^{-11/2} \approx 0.0221$ is the one selected by that sweep. The ROM keeps $K_m = 2$ eigenmodes ($K = 4$), so $K_f > K_m$ and the advection couples the ROM to the tail.

The advection is not folded into the damped-wave operator. $A_{\mathrm{full}} = (J_{\mathrm{full}} - R_{\mathrm{full}})P_{\mathrm{full}}$ stays the pure damped wave, so $Q = \gamma P$ holds (relative error 7.039×10^{-15}), $\lambda_{K+1} = 0.5$ is unchanged, and (H2a) holds by eigenmode invariance (Lemma 1, residual 6.348×10^{-13}). For the same reason the block $P_0^{-1}\Phi^\top P_{\mathrm{full}} A_{\mathrm{full}} P_\perp$ vanishes to round-off (1.27×10^{-12}). The linear cross-coupling sits in the nonlinearity, $F_{\mathrm{nl}}(x) = A_{\mathrm{adv}}x + F_{\sin}(x)$ with

$$A_{\mathrm{adv}} = \begin{bmatrix} 0 & 0 \\ -c\,W_f D_h W_f & 0 \end{bmatrix},$$

and its ROM–tail block $A_{\mathrm{adv},r,\perp} := P_0^{-1}\Phi^\top P_{\mathrm{full}} A_{\mathrm{adv}} P_\perp$ of Lemma 4 has norm $\|A_{\mathrm{adv},r,\perp}\|_2 = 0.106$ in R2 and 0.005301 in R1.

Two regimes are used. R2 has $c = 1$ ($\mathrm{Pe} = 20$) and is open-loop unstable. R1 has $c = 0.05$ ($\mathrm{Pe} = 1$). The sampling period of R2 is $T_s = 3 \times 10^{-4}$ with $\tau_A = 1.5 \times 10^{-5}$, $\tau_B = 3 \times 10^{-5}$. This is a trade-off forced by the active tail coupling: at the previous $T_s = 5 \times 10^{-4}$ the certificate refuses at $w^\star$, because $\delta_{\mathrm{total}} = 0.01821 > \delta_{\mathrm{cert}} = 0.01455$ and the authority condition (H6), (27), fails (Table V). R1 keeps $T_s = 0.002$. The short period is set by the hold term, not by the plant time scale: the slowest mode ($\omega_{\min} = 1.222$) is sampled 1.714×10^4 times per period $2\pi/\omega_{\min}$ in R2 (2571 in R1), and the hold term $\phi L_{\mathrm{hold}} T_s$ of (21), with $L_{\mathrm{hold}} = 44.04$ and $\mathrm{Lip}(L_f h) = 1.77$, is a share 0.4872 of δ_{total} in R2 (0.5596 in R1, where $L_{\mathrm{hold}} = 1.474$ and $\mathrm{Lip}(L_f h) = 0.7948$).

Every constant of Sections II–III is computed at run time in float64 by a function that names the equation it implements. A pytest AST scan fails if a float literal appears in the numerical modules, and a perturbation test fails if any certificate quantity does not depend on the configuration inputs. The certificate raises (refuses) unless (H0)–(H7), the timing order and the codesign window all hold. This single flag is the Theorem 1 hypothesis row of Table III. (H-dz) is checked separately and only for Proposition 3. The Path-A LUT is indexed in port space $v = g_r^\top P z$ by default. When $\mathcal{C}_{r,0} = \{\|z\|_P^2 \le E_{\mathrm{safe}}\}$ as in (15), its range is $D_v = 2\sqrt{E_{\mathrm{safe}}}\max_j \sqrt{(g_r^\top P g_r)_{jj}} = 0.4472$ by (29); the largest sampled value is 0.4472 (both to four significant digits). The modulus $\mathrm{Lip}(L_f h)$ that enters L_{hold} carries the factor of two of the gradient $\nabla_z(z^\top P z) = 2Pz$; it is cross-checked against finite-difference quotients of $L_f h$ at random pairs in $\mathcal{C}_{r,0}$, which must not exceed the certified modulus.

The full-order plant is integrated with an exponential integrator (ETD2RK in the lattice eigenbasis). The latch instants τ_A, τ_B are integration nodes, so the switching times are exact. The disturbance is the worst case of Remark 3 with $\epsilon_{\mathrm{adv}} = 0.002$. All seeds come from the configuration file. The suite runs 248 experiment assertions and 65 pytest tests on two CPU cores in 642.9 s, with a peak resident memory of 1193 MB (Table XXI).

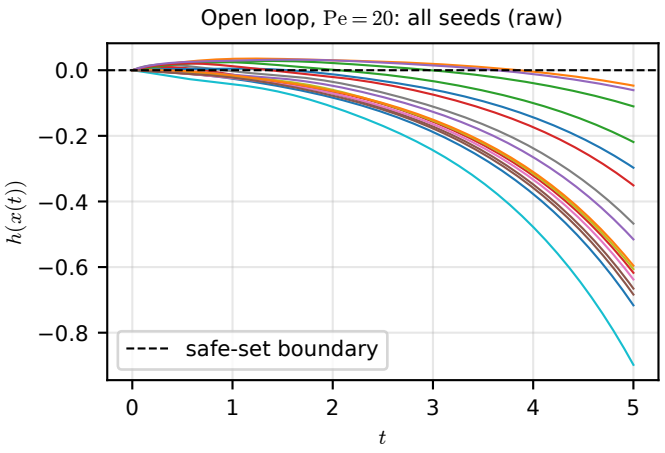


Fig. 1. Experiment 0: raw open-loop $h(x(t))$ in R2 ($\mathrm{Pe} = 20$) for all seeds.

B. Experiment 0: open-loop exit

With $\mathrm{Pe} = 20$ and $w = w^\star$ the linearisation has growth rate 0.2925 (Table I). All 16 of 16 initial conditions satisfying (H7) leave $\mathcal{C}$ without control: the worst $\min_t h$ is -0.8979 and the latest exit is at $t = 3.856$ (Fig. 1, Table IV). The safety problem is therefore not trivial: invariance in the later experiments comes from the controller, not from passivity.

C. Experiment 1: constants, hypotheses and refusals

Tables I and II list every certificate constant for R2 and R1, and Table III lists the hypotheses. Proposition 2 holds to round-off: $\mu_{2,P}(A_{r,0}) = -0.5$.

The tail coupling is now active. In R2, $\|A_{\mathrm{adv},r,\perp}\|_2 = 0.106$ and $L_{\perp,P} = 0.04741$. Of this, 0.04654 is the linear advective part and 8.719×10^{-4} the sin nonlinearity, so the advective part is a fraction 0.9816 of the total. In R1, $L_{\perp,P} = 0.003199$, and the ratio between the regimes is 14.82. The tail term $L_h L_{\perp,P}\varepsilon_{\mathrm{basis},P}$ is a share 0.2704 of $\delta_{\mathrm{total}} = 0.01356$ in R2 (and a share 0.006967 of $\delta_{\mathrm{total}} = 0.002634$ in R1).

TABLE I
CERTIFICATE CONSTANTS (PART 1 OF 2), COMPUTED AT RUN TIME IN FLOAT64 FROM THE LATTICE OPERATORS. K_f IS THE ADVECTION BAND, w THE DISCLOSED CUTOFF-MODE WEIGHT (TABLE VI); $L_{\perp,P}(\mathrm{R2})/L_{\perp,P}(\mathrm{R1}) = 14.82$.

quantity	equation	R2 (Pe = 20)	R1 (Pe = 1)
N_m	–	160	160
N	–	320	320
m	–	2	2
ν	–	0.05	0.05
α (plant)	–	1	1
γ	–	1	1
κ	–	0.05	0.05
c	–	1	0.05
K_m	–	2	2
K	–	4	4
full-weight advection modes	–	2	2
E_{safe}	(3)	0.05	0.05
$\varepsilon_{\mathrm{adv}}$	(7)	0.002	0.002
open-loop growth $\max \mathrm{Re}\,\lambda(A_{\mathrm{lin}})$	–	0.2925	-0.5
K_f	–	4	4
w (cutoff weight)	–	0.0221	0.0221
Pe	Remark 1	20	1
$\lambda_{\min}(P_{\mathrm{full},\epsilon})$	(2)	0.6892	0.6892
$\omega_{\min}$	(2)	1.222	1.222
$\mu_{2,P}(A_{r,0})$	(14)	-0.5	-0.5
$\|Q - \gamma P\|/\|\gamma P\|$	(14)	7.039×10^{-15}	7.039×10^{-15}
λ_{K+1}	(6)	0.5	0.5
(H2a) residual $\|\Phi^\top P_{\mathrm{full},\epsilon} P_\perp\|$	Lemma 1	6.348×10^{-13}	6.348×10^{-13}
$\|A_{r,\perp}\|_2$ (eigenmode invariance)	Lemma 1	1.27×10^{-12}	1.27×10^{-12}
ρ_{ODD}	(3)	0.2828	0.2828
D_{ODD}	(15)	0.5656	0.5656
$\mathrm{Lip}(F_{\mathrm{nl}})$	Assumption 1	2.67	0.1354
$M_\perp$	(7)	0.04916	0.003647
$\varepsilon_{\mathrm{basis},P}$	(9)	0.09831	0.007293
$\|A_{\mathrm{adv},r,\perp}\|_2$	Lemma 4	0.106	0.005301
$L_{\perp,P}$ (advective, linear)	(8)	0.04654	0.002327
$L_{\perp,P}$ (sin nonlinearity)	(8)	8.719×10^{-4}	8.719×10^{-4}
$L_{\perp,P}$	(8)	0.04741	0.003199
linear share of $L_{\perp,P}$	(8)	0.9816	0.7274
L_h	(16)	0.7865	0.7865
g_h	(16)	0.4788	0.4788
d_h	(16)	0.4472	0.4472
M_z	(16)	4.324	0.7887
$\mathrm{Lip}(L_f h)$	(16)	1.77	0.7948
$\mathrm{Lip}(L_g h)$	(16)	2.394	2.394
$L_{\mathcal{T},z}$	(10)	5.985	0.5985
$\eta_{\mathcal{T}}$	Lemma 2	4.975	0.4975
(H-sat) left side $L_{\mathcal{T},z}\sqrt{K} D_{\mathrm{ODD}}/2$	(11)	3.386	0.3386
$u_{\max}$	(11)	3.4	0.35
$\varepsilon_{\mathrm{CROWN}}$ (backward)	(12)	0.004455	4.536×10^{-4}

With $\varepsilon_{\mathrm{basis},P} = 0.09831$, the R2 margin is $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}} = 0.01456 - 0.01356 = 9.994 \times 10^{-4}$.

Lemma 4 gives $c_M = 0.3725$ from $\lambda_{\min}(M, P) = 0.3731$ and $s_{\mathrm{nl}} = 3.005 \times 10^{-4}$ in R2, and $c_M = 0.9829$ from $\lambda_{\min}(M, P) = 0.9835$ and $s_{\mathrm{nl}} = 3.005 \times 10^{-4}$ in R1. The codesign window is $[8, 11]$ and contains $b_A = 9$.

Table V shows the certificate refusing 6 inadmissible variants. The state-space LUT (the $m = K$ variant of Remark 2) has an empty window $[9, 5]$. The other refusals are $u_{\max}$ below the (H-sat) left side, unfiltered lattice advection, the plain projector Π_f, the previous sampling period and T_s, τ_A, τ_B scaled by 10. Proposition 3 needs (i) $\mu_R > L_{\mathrm{sec}}$, (H-dz) and (H3) (Table III). In R1 all three hold: $\mu_R = 0.5 > L_{\mathrm{sec}} = 0.1204$ and $r_{\mathrm{ss}} = 0.01162 < r_{\min} = 0.156$. In R2, $L_{\mathrm{sec}} = 2.392$ exceeds $\mu_R = 0.5$, so condition (i) fails and Proposition 3 does not apply; this does not affect Theorem 1.

D. Experiment 9: the cutoff-mode weight and the certifiable frontier

The weight w is swept over 17 geometric values from 1 down to 0.003906 at the R2 Péclet number (Table VI, Fig. 2). At every w the open loop leaves $\mathcal{C}$ on all seeds, so the plant is unsafe throughout the sweep. For $w \geq 0.08839$, the tail tolerance (H4) fails together with (H6) and (H7). For $0.03125 \leq w \leq 0.0625$, only the authority condition (H6) fails. The certificate accepts for $w \leq w^\star = 0.0221$. Below $w^\star$ the single-rate ablation (c) is no longer refused (first at $w = 0.01562$). The deployed $w^\star$ is therefore the largest grid value that is accepted with a positive authority margin $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}}$ in (27) while every ablation still breaks. $\|A_{\mathrm{adv},r,\perp}\|_2$ is proportional to w, and the tail share of δ_{total} falls from 0.9968 at $w = 1$ to 0.2704 at $w^\star$.

For the plain projector ($w = 1$), the Péclet scan of Ta-

TABLE II
CERTIFICATE CONSTANTS (PART 2 OF 2).

quantity	equation	R2 (Pe = 20)	R1 (Pe = 1)
E_q (explicit)	(13)	0.005674	5.58×10^{-4}
$\varepsilon_{\mathrm{CROWN}}$	Prop. 1	0.004455	4.536×10^{-4}
D_v	(29)	0.4472	0.4472
D_v (sampled sup)	(29)	0.4472	0.4472
M_v	(17)	5.176	0.9441
$\Delta_{q,A}$ (LUT step)	(29)	8.735×10^{-4}	8.735×10^{-4}
ε_A	(17)	0.007543	7.624×10^{-4}
Δu_{lag}	(17)	0.007765	9.441×10^{-4}
window lower bound	(18)	8	8
b_A	(18)	9	9
window upper bound	(18)	11	11
T_s	(5)	3×10^{-4}	0.002
τ_A	(5)	1.5×10^{-5}	1×10^{-4}
τ_B	(5)	3×10^{-5}	2×10^{-4}
samples per period $2\pi/(\omega_{\min} T_s)$	(5)	1.714×10^4	2571
b_B	Prop. 1	12	12
γ_0	(23)	0.35	0.3
w_1	(19)	0.05	0.04999
w_2	(19)	0.05	0.04999
w_3	(19)	0.9	0.9
$\phi(\gamma_0 T_s)$	(21)	0.5	0.5
L_{hold}	(20)	44.04	1.474
Δh_{dip}	(20)	2.06×10^{-6}	3.008×10^{-6}
δ_{total}: tail term	(21)	0.003666	1.835×10^{-5}
δ_{total}: quantisation term	(21)	0.002207	2.246×10^{-4}
δ_{total}: lag term	(21)	1.859×10^{-4}	2.26×10^{-5}
δ_{total}: adversary term	(21)	8.944×10^{-4}	8.944×10^{-4}
δ_{total}: hold term	(21)	0.006607	0.001474
δ_{total}	(21)	0.01356	0.002634
tail share $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P} / \delta_{\mathrm{total}}$	(21)	0.2704	0.006967
hold share of δ_{total}	(21)	0.4872	0.5596
h_{act}	(22)	0.02068	0.02559
$r_{\min}$	(22)	0.1402	0.156
$\lambda_{\min}(M, P)$	Lemma 4	0.3731	0.9835
s_{nl}	Lemma 4	3.005×10^{-4}	3.005×10^{-4}
c_M	Lemma 4	0.3725	0.9829
δ_{cert}	Lemma 4	0.01456	0.03161
δ_{cert} branch	(30)	1	1
margin $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}}$	(27)	9.994×10^{-4}	0.02898
$\gamma_0 h_{\mathrm{act}}$ (Cor. 1)	Cor. 1	0.007237	0.007678
null-space rate	(24)	1	1
μ_R	Prop. 3	0.5	0.5
L_{sec}	Prop. 3	2.392	0.1204
$\bar{\varepsilon}_q$	Prop. 3	0.01998	0.002342
r_{ss}	(26)	∞	0.01162

ble VII shows that the certifiable advection speeds ($c \leq 0.02$, $\mathrm{Pe} \leq 0.4$) and the open-loop-unstable speeds ($c \geq 0.5$, $\mathrm{Pe} \geq 10$) are disjoint. Under the Euclidean-ODD supremum that defines $M_\perp$ in (7), full-strength cutoff-mode advection is not certifiable at any Péclet number where the open loop is unstable: no softening. The plain-projector result is the L^2-ODD conservatism of Remark 1. Keeping A_{adv} inside F_{nl} preserves (H2a) exactly (eigenmode invariance, Lemma 1) at the price of charging $A_{\mathrm{adv},r,\perp}$ in $M_\perp$ and $L_{\perp,P}$. The weight w attenuates the cutoff modes, which is what an H^1-bounded ODD would do for a smooth field; certifying $w = 1$ at $\mathrm{Pe} \geq 10$ would require that H^1 formulation, which is outside the present theorem.

E. Experiment 2: Proposition 1 (twin-CROWN)

For $b_B = 3, \ldots, 14$, with nearest and Adam-learned rounding, the twin-CROWN bound exceeds the largest Student–Teacher gap over 100016 samples (uniform samples of $\mathcal{Z}_{\mathrm{ODD}}$ plus all box vertices) by a factor of 1.887–2.844 (Fig. 3, Table VIII). The bound is monotone non-increasing in b_B and halves with every additional bit of b_B. At $b_B = 12$, learned rounding lowers $\varepsilon_{\mathrm{CROWN}}$ from 0.004718 to 0.004455. The port box that indexes the Path-A table is certified as well.

F. Experiment 3: δ_{cert}, Lemma 4 and Lemma 3

In class (H), the certified bound of Lemma 4 is compared with the brute-force minimum of the exact bracket of (23) over samples of $\mathcal{C}_{r,\mathrm{act}}$. The bound is $0.01456 \leq 0.01468$ in R2 and $0.03161 \leq 0.03164$ in R1, so it is sound and nearly tight. In both regimes $\delta_{\mathrm{total}} \leq \delta_{\mathrm{cert}}$ (Fig. 4).

On a class-(D) instance, the closed form (25) equals the brute-force infimum of its affine bound in both branches and lies below the exact-bracket minimum (Table IX). This is a standalone check of the lemma, so δ_{total} is reported as N/A

TABLE III
HYPOTHESIS LEDGER OF THEOREM 1, HYPOTHESES (H0)–(H7), AND OF PROPOSITION 3, CONDITIONS (I) $\mu_R > L_{\text{sec}}$ AND (H-DZ); ✓ = SATISFIED.

hypothesis	content	R2	R1
(H0)	ODD containment and E_{safe}	✓	✓
(H1)	Assumption 1 constants on $\mathcal{X}_{\text{ODD}}$	✓	✓
(H2)	(H2a)–(H2c), $\gamma < 2\omega_{\min}$, $\epsilon = \gamma/2$	✓	✓
(H3)	Teacher: $H/2 \geq m$, $G_{\text{port}} \succ 0$, (H-sat)	✓	✓
(H4)	tail tolerance, h_{act} interval	✓	✓
(H5)	$b_A \geq 1$ satisfies (4)	✓	✓
(H6)	authority (27)	✓	✓
(H7)	initial conditions	✓	✓
(H-sat)	$L_{\mathcal{T},z}\sqrt{K}D_{\text{ODD}}/2 \leq u_{\max}$ (11)	✓	✓
timing $0 < \tau_A < \tau_B < T_s$	latch order of (5)	✓	✓
codesign window (18)	b_A inside the window	✓	✓
Thm. 1 hypotheses	all of the above	✓	✓
$\mu_R > L_{\text{sec}}$	Prop. 3(i) contraction rate	×	✓
(H-dz)	$r_{\text{ss}} < r_{\min}$ (Prop. 3 only)	×	✓
Prop. 3 hypotheses	(i) $\mu_R > L_{\text{sec}}$, (H-dz) and (H3)	×	✓

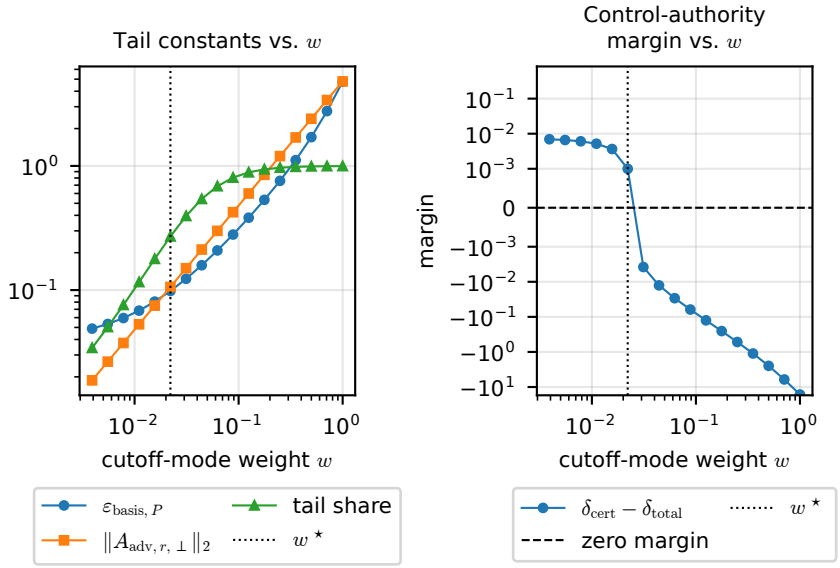


Fig. 2. Experiment 9: tail constants (left) and the authority margin $\delta_{\text{cert}} - \delta_{\text{total}}$ of (27) (right) versus the cutoff-mode weight w; the dotted line marks $w^\star$.

there. Using branch 1 unconditionally would overestimate the margin when $c < \gamma_0$.

G. Experiment 4: Theorem 1 (positive case)

The full method runs with the worst-case adversary on 16 seeds per regime from two kinds of initial condition: the boundary start $\tilde{h}_r(z(0)) = 0$, with the tail at its largest admissible value (H7), and the interior start $h_r(z(0)) = 0.5\,h_{\text{act}}$. Conclusions (a)–(c) hold at every integrator node for both starts and both regimes (Fig. 5, Table X); the largest tail ratio $\|x_\perp\|/\varepsilon_{\text{basis},P}$ (norm weighted by $P_{\text{full},\epsilon}$) is 1 in R2 and 1 in R1. For both starts the minimum of $h(x(t))$ over the run is attained at the initial instant ($t = 0$ for the boundary start, $t = 0$ for the interior start): $\min_t h$ is 2.06×10^{-6} in R2 and 3.008×10^{-6} in R1 from the boundary, 0.01034 and 0.0128 from the interior. The inter-sample dip floor $-\Delta h_{\text{dip}}$ of conclusion (b) is therefore not exercised dynamically by these runs. Experiment 4b measures how much of each budget term the same trajectories use.

H. Experiment 4b: budget utilisation along the certified trajectories

Experiment 4b re-runs the closed loops of Experiment 4 and evaluates, at each of the 1600032 (R2) and 240000 (R1) integrator nodes over both starts and all seeds, every term of (1) next to the bound the proof uses for it (Table XI). Each fraction below is the realised term over its bound: the maximum over all nodes and, in parentheses, the median over the 102636 (R2) and 20238 (R1) nodes whose sample z_k lies in $\mathcal{C}_{r,\text{act}}$, where the budget enters the proof. A fraction 0.9359 (R2) and 0.9157 (R1) of the nodes belong to periods with $z_k \in \mathcal{B}_{\text{int}}$; there the proof uses only the Lipschitz branch, and its inequality holds at every such node.

- Adversary: 0.8772 (0.1454) in R2 and 0.9761 (0.2814) in R1. The worst-case w has norm ε_{adv} at every node, so the gap to one is the supremum of $\|d_r^\top \nabla \tilde{h}_r\|_2$ over $\mathcal{C}_{r,0}$ in $\overline{d}_h$.
- Quantization and lag: 0.3729 (0.01712) and 0.3672 (0.03416). The mismatch e_u reaches at most 0.5872 (R2)

TABLE IV
EXPERIMENT 0: OPEN-LOOP ($u \equiv 0$) TRAJECTORIES OF THE R2 BENCHMARK (PE = 20) FROM CERTIFIED INITIAL STATES.

seed	$h(x_0)$	$\min_t h(x(t))$	first exit time
0	2.06×10^{-6}	-0.7165	3×10^{-4}
1	2.06×10^{-6}	-0.04703	3.856
2	2.06×10^{-6}	-0.1102	2.983
3	2.06×10^{-6}	-0.6174	0.7137
4	2.06×10^{-6}	-0.5155	0.4689
5	2.06×10^{-6}	-0.6655	0.0657
6	2.06×10^{-6}	-0.6377	9×10^{-4}
7	2.06×10^{-6}	-0.4675	0.8838
8	2.06×10^{-6}	-0.606	0.4128
9	2.06×10^{-6}	-0.8979	3×10^{-4}
10	2.06×10^{-6}	-0.2972	1.566
11	2.06×10^{-6}	-0.5958	0.2889
12	2.06×10^{-6}	-0.2189	2.061
13	2.06×10^{-6}	-0.351	1.379
14	2.06×10^{-6}	-0.06081	3.636
15	2.06×10^{-6}	-0.6837	3×10^{-4}

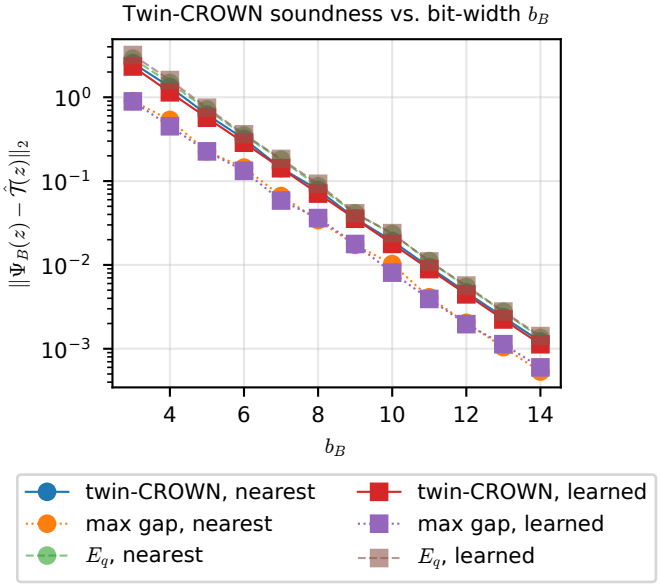


Fig. 3. Experiment 2: certified $\varepsilon_{\mathrm{CROWN}}$, explicit E_q and the largest sampled gap versus b_B.

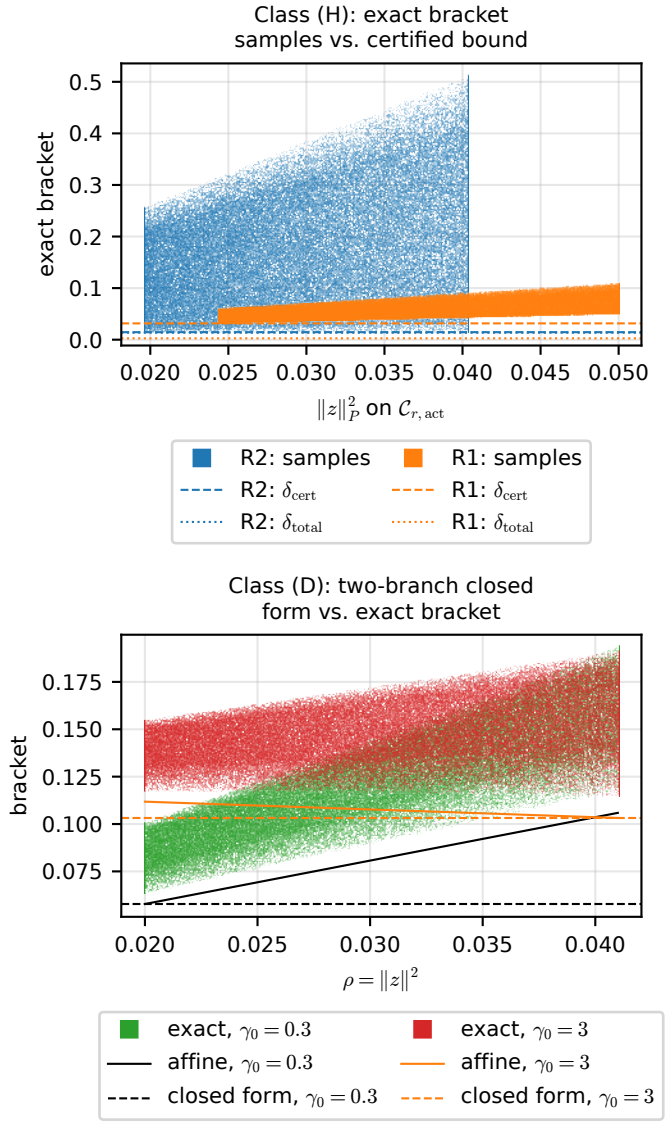


Fig. 4. Experiment 3: exact bracket samples versus the certified δ_{cert} and δ_{total} (top); class-(D) closed form versus the exact bracket (bottom).

and 0.5705 (R1) of its phase bound, so $\varepsilon_{\mathrm{CROWN}}$, ε_A and Δu_{lag} are the conservative constants of this term.

- Tail coupling: 0.3161 (0.003444) and 0.2606 (0.01455). Of the factors of $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P}$, the two least used are the Lipschitz constant $L_{\perp,P}$ (median $\|\Delta_\perp\|_2/(L_{\perp,P}\|x_\perp\|) = 0.2554$ in R2 and 0.1842 in R1) and the alignment of $\nabla\tilde{h}_r$ with $\Delta_\perp$ (median cosine 0.2113 and 0.2817).
- Hold shift: 0.07439 (0.001438) and 0.09728 (0.0226), the smallest of the four fractions in both regimes. Two constants of L_{hold} are conservative here: $\|z - z_k\|_2$ reaches at most 0.3205 (R2) and 0.5853 (R1) of $M_z s$, and the hold shift at most 0.2323 (R2) and 0.1794 (R1) of $L_{\mathrm{hold}}\|z - z_k\|_2/M_z$, its Lipschitz bound in z.

The realised rate $\dot{\tilde{h}}_r + \gamma_0 \tilde{h}_r$ minus the right-hand side of (1) is positive at every active-shell node, so the soundness check passes. Its minimum is $0.6597\,\delta_{\mathrm{total}}$ in R2 and $11.5\,\delta_{\mathrm{total}}$ in R1, attained at a sampling instant, where the hold term vanishes; the Teacher margin at z_k (the left-hand side of (23) minus δ_{total}) supplies a fraction 0.5084 (R2) and 0.9691 (R1) of it.

I. Experiment 5: Proposition 3 and ablation (d)

In R1, (H-dz) holds and the trajectories enter $\mathcal{B}_{\mathrm{ss}}$. With the mid-tread deadzone, Path A is silent late in the run ($u_A \equiv 0$) for every $b_A \in \{3, 5, 7, 9\}$. Ablation (d) replaces it with a mid-rise quantiser and gives a persistent actuator limit cycle at every b_A, including the deployed $b_A = 9$ (Figs. 6 and 7, Table XII). The number of sign switches per unit time falls from 953.1 at $b_A = 3$ to 121.6 at $b_A = 9$, and the final lag error $\|u_A - \hat{\mathcal{T}}\|_2$ of the mid-rise quantiser at $b_A = 9$ is 3.507×10^{-4}. With $b_A = 1$, $r_{\mathrm{ss}} = 0.3992 > r_{\min} = 0.156$, so (H-dz) fails and Proposition 3 is refused.

TABLE V
CERTIFICATE REFUSALS ON NON-ADMISSIBLE VARIANTS OF R2. THE LUT WINDOW IS (18) IN THE VARIANT'S OWN FORM (PORT SPACE BY DEFAULT; THE STATE-SPACE ROW USES K, D_{ODD} AND M_z).

configuration	refused	failing	δ_{total}	δ_{cert}	window [lo, hi]
state-space LUT ($m = K$ variant)	✓	(H5), codesign window (18)	0.01377	0.01456	[9, 5]
$u_{\max}$ below the (H-sat) left side	✓	(H3)	0.01245	0.01483	[8, 11]
pure lattice advection (no spectral cutoff)	✓	(H4), (H6), (H7)	157.6	−3091	[3, 11]
plain projector Π_f ($w = 1$)	✓	(H4), (H6), (H7)	7.93	−8.456	[6, 11]
previous sampling period $T_s = 5 \times 10^{-4}$ at $w^\star$	✓	(H6)	0.01821	0.01455	[7, 11]
T_s, τ_A, τ_B ×10 (hold term)	✓	(H6)	0.07483	0.01438	[5, 11]

TABLE VI
DISCLOSED SWEEP OF THE CUTOFF-MODE ADVECTION WEIGHT w AT THE R2 PÉCLET NUMBER ($w = 1$: PLAIN PROJECTOR Π_f). THE DEPLOYED $w^\star$ IS THE LARGEST GRID VALUE ACCEPTED WITH A POSITIVE AUTHORITY MARGIN IN (27) AND ALL ABLATIONS REFUSED. TAIL SHARE: $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P} / \delta_{\mathrm{total}}$; MARGIN: $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}}$; EXITS: OPEN-LOOP EXITS OF 16 SEEDS AT $c = 1$; –: ABLATIONS NOT EVALUATED (THE FULL CERTIFICATE ALREADY REFUSES).

w	Thm. 1 hypotheses	failing	$\varepsilon_{\mathrm{basis},P}$	tail share	$\|A_{\mathrm{adv},r,\perp}\|_2$	margin	abl. refused	exits
1	×	(H4), (H6), (H7)	4.77	0.9968	4.798	−16.39	–	16/16
0.7071	×	(H4), (H6), (H7)	2.766	0.995	3.393	−6.088	–	16/16
0.5	×	(H4), (H6), (H7)	1.706	0.9911	2.399	−2.492	–	16/16
0.3536	×	(H4), (H6), (H7)	1.113	0.9831	1.696	−1.107	–	16/16
0.25	×	(H4), (H6), (H7)	0.7587	0.9677	1.2	−0.5211	–	16/16
0.1768	×	(H4), (H6), (H7)	0.5331	0.9388	0.8482	−0.2539	–	16/16
0.125	×	(H4), (H6), (H7)	0.3829	0.8879	0.5998	−0.1256	–	16/16
0.08839	×	(H4), (H6), (H7)	0.2801	0.8052	0.4241	−0.06177	–	16/16
0.0625	×	(H6)	0.2086	0.6865	0.2999	−0.02934	–	16/16
0.04419	×	(H6)	0.1585	0.5417	0.212	−0.01265	–	16/16
0.03125	×	(H6)	0.1232	0.395	0.1499	−0.003795	–	16/16
0.0221	✓	none	0.09831	0.2704	0.106	9.994×10^{-4}	✓	16/16
0.01562	✓	none	0.08073	0.1782	0.07497	0.003661	×	16/16
0.01105	✓	none	0.0683	0.1159	0.05301	0.00518	×	16/16
0.007812	✓	none	0.05952	0.0758	0.03748	0.006072	×	16/16
0.005524	✓	none	0.0533	0.05036	0.02651	0.006613	×	16/16
0.003906	✓	none	0.04891	0.03419	0.01874	0.006949	×	16/16

TABLE VII
PLAIN PROJECTOR Π_f ($w = 1$): CERTIFICATE ACCEPTANCE VERSUS OPEN-LOOP INSTABILITY OVER THE ADVECTION SPEED c. THE TWO SETS ARE DISJOINT.

c	Pe	Thm. 1 hypotheses	failing	$\varepsilon_{\mathrm{basis},P}$	tail share $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P} / \delta_{\mathrm{total}}$	max Re eig$(A_{\mathrm{full}} + A_{\mathrm{adv}})$
0.005	0.1	✓	none	0.06197	0.05895	−0.5
0.01	0.2	✓	none	0.08563	0.1427	−0.5
0.02	0.4	✓	none	0.1329	0.3361	−0.5
0.03	0.6	×	(H6)	0.1803	0.5053	−0.5
0.05	1	×	(H4), (H6), (H7)	0.2749	0.7202	−0.5
0.1	2	×	(H4), (H6), (H7)	0.5114	0.9039	−0.5
0.2	4	×	(H4), (H6), (H7)	0.9846	0.9716	−0.5
0.3	6	×	(H4), (H6), (H7)	1.458	0.9859	−0.365
0.5	10	×	(H4), (H6), (H7)	2.404	0.9935	0.1025
1	20	×	(H4), (H6), (H7)	4.77	0.9968	0.8887

R1 is open-loop stable, so the chattering appears in the actuator signal, not as a larger state amplitude.

J. Experiment 6: ablations (a), (b), (c), (e)

Each ablation removes one component at the deployed $w^\star$. In every case the full certificate refuses and the full-order plant leaves $\mathcal{C}$ (Fig. 8, Table XIII). For (a) and (b), a naive certificate with the corresponding term dropped would accept.

- (a) $K_m = 1$ without tail bookkeeping: 4 of 16 seeds exit.
- (b) $b_B = 2$ without the CROWN margin: 9 of 16 seeds exit.
- (c) Single-rate hold (Table XIV). At the deployed T_s, δ_{total} rises from 0.01356 to $0.01702 > \delta_{\mathrm{cert}}$, so (H6) fails. At $T_s = 0.2$, single rate gives 10 exits and multi-rate gives none.
- (e) $\epsilon_{\mathrm{adv}} = 1$: 16 of 16 seeds exit. At the largest certifiable $\epsilon^\star_{\mathrm{adv}} = 0.004223$ there are no exits.

K. Experiment 7: Corollary 1 and ablation (f)

In R1 the paths start on the level 0.9 of $\tilde{\mathcal{C}}_r$ ($h(x_0) = 0.005051$), with spatially smooth noise. The intensity is chosen by a grid search over $\sigma \in \{0.002, 0.003, 0.005, 0.007, 0.01, 0.014, 0.02\}$ (Table XV). $\sigma^\star = 0.007$ is the largest grid value for which the Corollary accepts at $T = 2$ and

TABLE VIII
PROPOSITION 1: TWIN-CROWN BOUND VS. THE SAMPLED STUDENT–TEACHER GAPS ON 100016 POINTS OF $\mathcal{Z}_{\mathrm{ODD}}$ (100000 UNIFORM SAMPLES PLUS ALL 16 BOX VERTICES).

b_B	rounding	$\varepsilon_{\mathrm{CROWN}}$	E_q (explicit)	max sampled gap	mean gap	bound / max gap
3	nearest	2.559	2.887	0.9001	0.3945	2.844
3	learned	2.326	3.189	0.8927	0.3574	2.606
4	nearest	1.329	1.481	0.5366	0.1701	2.476
4	learned	1.145	1.603	0.4509	0.1397	2.54
5	nearest	0.6217	0.7167	0.2263	0.07976	2.747
5	learned	0.5696	0.7491	0.2261	0.0715	2.519
6	nearest	0.319	0.3508	0.1449	0.03818	2.201
6	learned	0.2868	0.3616	0.1326	0.03538	2.162
7	nearest	0.143	0.1801	0.0664	0.01819	2.153
7	learned	0.1424	0.1838	0.0586	0.01816	2.43
8	nearest	0.07865	0.08686	0.03422	0.00951	2.298
8	learned	0.07095	0.09282	0.03621	0.00937	1.96
9	nearest	0.03591	0.04118	0.01733	0.004777	2.072
9	learned	0.03554	0.04131	0.01776	0.00476	2.001
10	nearest	0.01931	0.02345	0.01023	0.002453	1.888
10	learned	0.01781	0.02394	0.00809	0.002398	2.202
11	nearest	0.009407	0.01112	0.004111	0.001221	2.288
11	learned	0.008911	0.01101	0.003928	0.001202	2.268
12	nearest	0.004718	0.005478	0.002039	6.126×10^{-4}	2.314
12	learned	0.004455	0.005674	0.001966	6.014×10^{-4}	2.266
13	nearest	0.002385	0.002768	0.001043	3.084×10^{-4}	2.287
13	learned	0.002235	0.002777	0.001134	3.016×10^{-4}	1.97
14	nearest	0.001224	0.001345	5.346×10^{-4}	1.568×10^{-4}	2.29
14	learned	0.001131	0.001417	5.994×10^{-4}	1.509×10^{-4}	1.887

TABLE IX
CERTIFIED MARGIN δ_{cert}: CLASS-(H) BOUND OF LEMMA 4 ($c_M = \lambda_{\min}(M,P) - 2s_{\mathrm{nl}}$, TWO-BRANCH INFIMUM) AND THE CLASS-(D) CLOSED FORM (25) VS. THE BRUTE-FORCE INFIMUM OF THE EXACT BRACKET OVER $\mathcal{C}_{r,\mathrm{act}}$. FOR CLASS (D) THE COLUMN c_M HOLDS THE CLOSED-FORM SLOPE c.

case	bound	c_M	$\lambda_{\min}(M,P)$	s_{nl}	γ_0	branch	δ_{cert} (bound)	brute-force inf	δ_{total}
R2	Lemma 4	0.3725	0.3731	3.005×10^{-4}	0.35	1	0.01456	0.01468	0.01356
R1	Lemma 4	0.9829	0.9835	3.005×10^{-4}	0.3	1	0.03161	0.03164	0.002634
class (D)	(25)	2.59	–	–	0.3	1	0.0578	0.06345	N/A (standalone)
class (D)	(25)	2.59	–	–	3	2	0.1032	0.1148	N/A (standalone)

TABLE X
THEOREM 1: MEASURED MINIMA OVER ALL SEEDS AND INTEGRATOR NODES FOR THE BOUNDARY START ($\tilde{h}_r(z(0)) = 0$) AND THE INTERIOR START ($h_r(z(0)) = 0.5\, h_{\mathrm{act}}$), WORST-CASE ADVERSARY.

config	start	seeds	$\min_t h(x_t)$	time of min	$\min \tilde{h}_r$	max tail / $\varepsilon_{\mathrm{basis},P}$	margin $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}}$
R2	boundary	16	2.06×10^{-6}	0	-7.294×10^{-17}	1	9.994×10^{-4}
R2	interior	16	0.01034	0	6.709×10^{-4}	0.0583	9.994×10^{-4}
R1	boundary	16	3.008×10^{-6}	0	-7.774×10^{-17}	1	0.02898
R1	interior	16	0.0128	0	0.01274	0.04749	0.02898

$\max_t |h(x_t) - h(x_0)| \geq 0.1\, h(x_0)$ holds on at least half of the paths; the Corollary refuses from $\sigma = 0.01$ on (failing condition: margin). On the screening paths at $\sigma^\star$ the criterion is driven mostly by the closed-loop drift (median deviation $8.41\, h(x_0)$). The noise itself is not negligible: the median deviation from the noise-free trajectory is $0.174\, h(x_0)$ (Table XV).

Out of 2000 Monte Carlo paths, 0 exit. The exact Clopper–Pearson 95% upper limit is $0.001843 \leq \alpha = 0.05$ (Fig. 9, Table XVI). The bound is very conservative: $T_{\mathrm{TTL}} = 2022$ exceeds the horizon by a factor of 1011, and no exit is observed.

Ablation (f) uses spatially white small-scale noise. The full Corollary refuses it: the tail envelope $\Delta_\sigma = 1.1$ exceeds $\sqrt{E_{\mathrm{safe}}} = 0.2236$. A cruder bound that ignores the tail diffusion ($\bar{\sigma}_\perp := 0$) accepts it, yet 2000 of 2000 paths exit (Clopper–Pearson lower limit $0.9982 > \alpha$). The tail-diffusion term is therefore load-bearing. The constants are split into Tables XVII and XVIII.

The Corollary is not exercised on R2. There the pointwise penalty $\gamma_0 h_{\mathrm{act}} = 0.007237$ already exceeds the margin $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}} = 9.994 \times 10^{-4}$, so its authority condition fails even at $\sigma = 0$ (asserted; Table II). All stochastic runs use $r_{W_2} = 0$, i.e. no distributional drift of the adversary.

L. Experiment 8: numerical method

At the convergence-test step, explicit Euler is unstable (amplification 3.73). The stiffness is moderate (spectral radius 72,

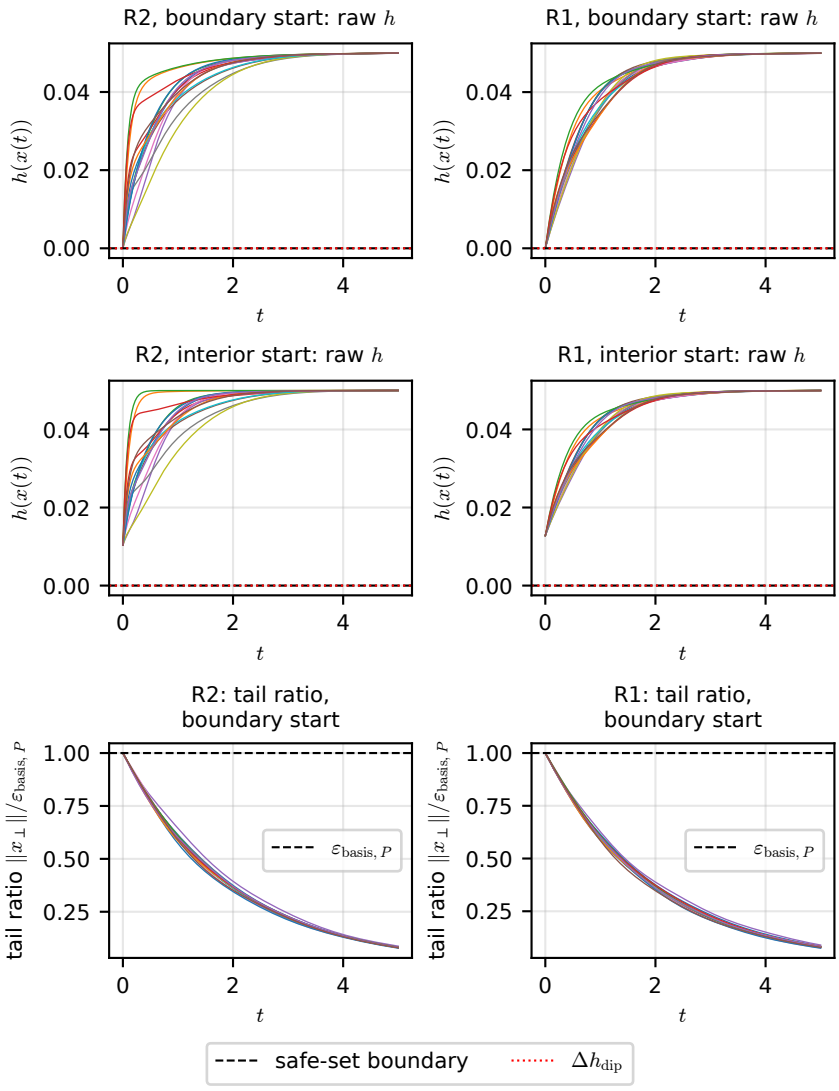


Fig. 5. Experiment 4: raw closed-loop $h(x(t))$ for the boundary start (top row) and the interior start (middle row), and the tail ratio $\|x_\perp\|/\varepsilon_{\text{basis},P}$, with the norm weighted by $P_{\text{full},\epsilon}$, for the boundary start (bottom row); R2 left, R1 right.

stiffness ratio 43.83). Step halving gives error ratios of 4.002–4.045 for ETD2RK, i.e. order 2 (Fig. 10, Table XIX). The simulator matches the closed-form augmented exponential of a latched linear system to 2.808×10^{-14}, whereas rounding the latch instants to a step grid gives 0.01675. The sparse modal propagation equals a dense nodal ETD2RK to 1.928×10^{-14}, and two runs give identical SHA-256 hashes (Table XX).

TABLE XI

EXPERIMENT 4B: REALISED FRACTION OF EACH BUDGET TERM OF (1) ALONG THE CERTIFIED TRAJECTORIES OF EXPERIMENT 4 (BOTH STARTS, ALL SEEDS, EVERY INTEGRATOR NODE, WORST-CASE ADVERSARY); ROW (V): MINIMUM AND MEDIAN OF THE REALISED RATE MINUS THE RIGHT-HAND SIDE OF (1), IN UNITS OF δ_{total}. COLUMNS $\mathcal{C}_{r,\text{act}}$: NODES OF PERIODS WHOSE SAMPLE z_k LIES IN THE ACTIVE SHELL, WHERE THE BUDGET ENTERS THE PROOF.

config	term	max, all nodes	median, all nodes	max, $\mathcal{C}_{r,\text{act}}$	median, $\mathcal{C}_{r,\text{act}}$
R2	(i) tail coupling	0.3161	2.063×10^{-4}	0.3161	0.003444
R2	(ii) adversary	0.8772	0.01896	0.8772	0.1454
R2	(iii) quantization/lag	0.3729	0.001841	0.3729	0.01712
R2	(iv) hold shift	0.07439	5.455×10^{-6}	0.07439	0.001438
R2	(iii) phase $[0, \tau_A)$	0.2057	7.39×10^{-4}	0.2057	0.008388
R2	(iii) phase $[\tau_A, \tau_B)$	0.3729	0.003766	0.3729	0.0348
R2	(iii) phase $[\tau_B, T_s)$	0.2551	0.002002	0.2551	0.01809
R2	(v) slack / δ_{total}: min, median	N/A	N/A	0.6597	2.807
R1	(i) tail coupling	0.2606	4.582×10^{-4}	0.2606	0.01455
R1	(ii) adversary	0.9761	0.04194	0.9761	0.2814
R1	(iii) quantization/lag	0.3672	0.005189	0.3672	0.03416
R1	(iv) hold shift	0.09728	8.842×10^{-4}	0.09728	0.0226
R1	(iii) phase $[0, \tau_A)$	0.2559	0.002249	0.2559	0.02988
R1	(iii) phase $[\tau_A, \tau_B)$	0.3672	0.009477	0.3672	0.05098
R1	(iii) phase $[\tau_B, T_s)$	0.2856	0.005166	0.2856	0.02683
R1	(v) slack / δ_{total}: min, median	N/A	N/A	11.5	17.56

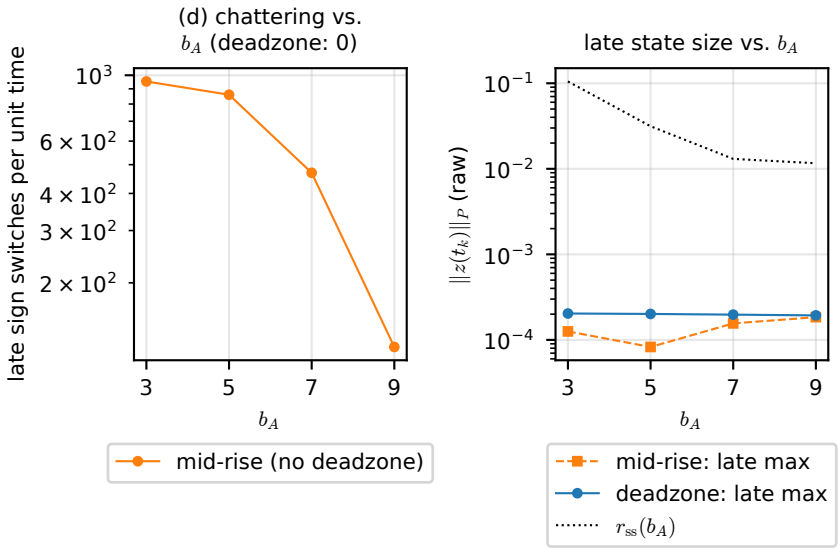


Fig. 6. Experiment 5 / ablation (d): late sign switches of the mid-rise Path-A command (left) and the late-window state size (right) versus b_A.

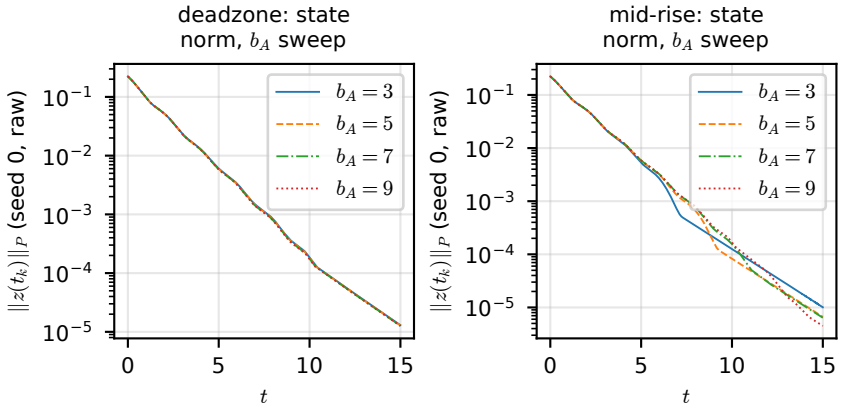


Fig. 7. Experiment 5: $\|z(t_k)\|_P$ for the deadzone (left) and mid-rise (right) quantiser over the b_A sweep.

TABLE XII

PROPOSITION 3 AND ABLATION (D) ON R1 (DISTURBANCE $w \equiv 0$): MID-TREAD LUT WITH DEADZONE VS. MID-RISE LUT (NO ZERO LEVEL) AT THE DEPLOYED $b_A = 9$ AND OVER THE SWEEP $b_A \in \{3, 5, 7, 9\}$; THE LAST ROW IS $b_A = 1$, WHICH VIOLATES (H-DZ). FINAL LAG: min OVER SEEDS OF $\|u_A(t_k) - \hat{\mathcal{T}}(z(t_k))\|_2$ AT THE LAST SAMPLE. FOR EVERY SWEPT b_A, $u_A \equiv 0$ IN THE LATE WINDOW WITH THE DEADZONE AND $u_A \neq 0$ AT EVERY LATE SAMPLE WITHOUT IT (ASSERTED).

LUT	b_A	ε_A	r_{ss}	$r_{\min}$	(H-dz)	late max $\|z\|_P$	final lag $\|u_A - \hat{\mathcal{T}}\|_2$	switches/time	min h
deadzone	3	0.02022	0.105	0.156	✓	2.035×10^{-4}	1.966×10^{-6}	0	3.939×10^{-6}
deadzone	5	0.005395	0.03145	0.156	✓	2.011×10^{-4}	1.914×10^{-6}	0	3.939×10^{-6}
deadzone	7	0.001689	0.01306	0.156	✓	1.974×10^{-4}	1.854×10^{-6}	0	3.939×10^{-6}
deadzone	9	7.624×10^{-4}	0.01162	0.156	✓	1.934×10^{-4}	1.833×10^{-6}	0	3.939×10^{-6}
mid-rise	3	0.02022	0.105	0.156	✓	1.257×10^{-4}	0.0197	953.1	3.939×10^{-6}
mid-rise	5	0.005395	0.03145	0.156	✓	8.268×10^{-5}	0.004851	859.9	3.939×10^{-6}
mid-rise	7	0.001689	0.01306	0.156	✓	1.559×10^{-4}	0.001286	469.4	3.939×10^{-6}
mid-rise	9	7.624×10^{-4}	0.01162	0.156	✓	1.845×10^{-4}	3.507×10^{-4}	121.6	3.939×10^{-6}
deadzone	1	0.07951	0.3992	0.156	×	2.067×10^{-4}	2.005×10^{-6}	0	3.939×10^{-6}

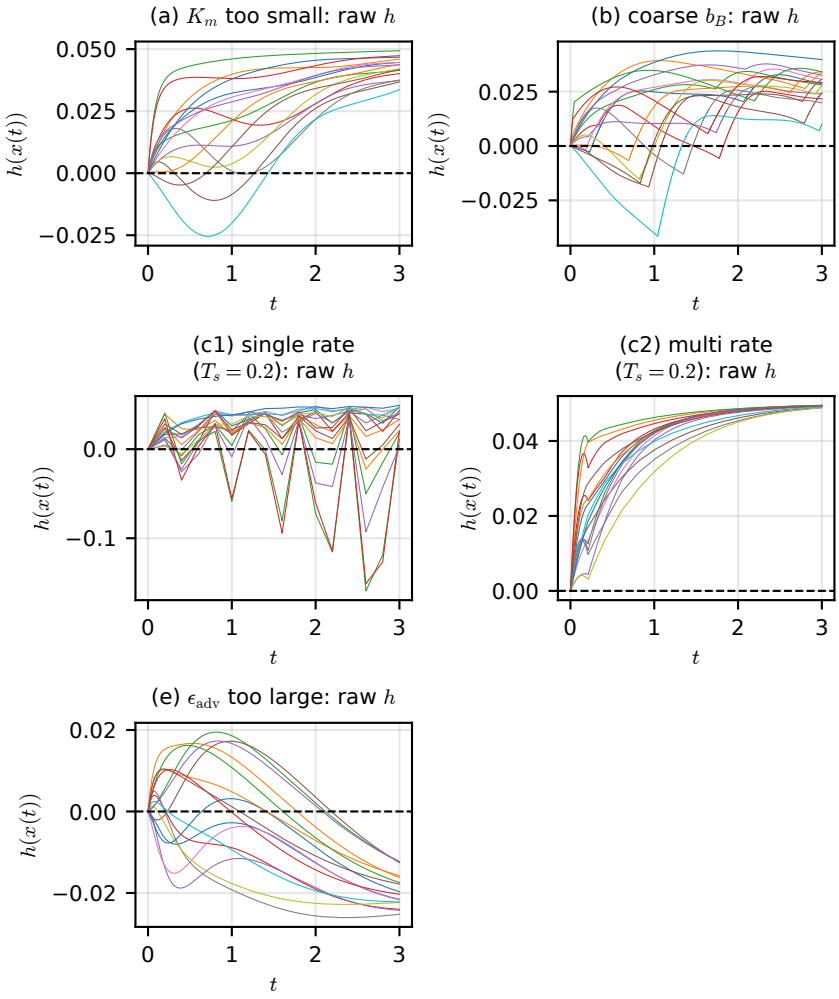


Fig. 8. Experiment 6: raw $h(x(t))$ under each ablation; violations appear below $h = 0$.

TABLE XIII

ABLATIONS AT THE DEPLOYED $w^\star$: EVERY REMOVED COMPONENT YIELDS A CERTIFICATE REFUSAL AND A FULL-ORDER SAFETY VIOLATION UNDER THE WORST-CASE ADVERSARY. ROW (E): THE LARGEST CERTIFIABLE ADVERSARY LEVEL IS $\epsilon^\star_{\text{adv}} = 0.004223$.

ablation	refused	failing	δ_{total}	δ_{cert}	naive δ_{total}	naive δ_{cert}	exits (of 16)	$\min_t h$
(a) $K_m = 1$, no tail bookkeeping	✓	(H3), (H4), (H6), (H7)	7.593	−60.63	0.007956	0.0335	4	−0.02545
(b) $b_B = 2$, no CROWN margin	✓	(H6)	2.149	0.01456	0.01135	0.01456	9	−0.04155
(c) single rate $\tau_A = \tau_B = T_s$	✓	(H6)	0.01702	0.01456	–	–	10	−0.1591
(e) $\epsilon_{\text{adv}} = 1$	✓	(H6)	0.4617	0.01409	–	–	16	−0.02605

TABLE XIV
ABLATION (C): EXPONENTIAL WEIGHTS w_1, w_2, Δh_{dip} AND δ_{total} FOR MULTI-RATE VS. SINGLE-RATE HOLD.

schedule	τ_A	τ_B	w_1	w_2	Δh_{dip}	δ_{total}: lag term	δ_{total}	(H6)
multi rate	1.5×10^{-5}	3×10^{-5}	0.05	0.05	2.06×10^{-6}	1.859×10^{-4}	0.01356	✓
single rate	3×10^{-4}	3×10^{-4}	1	0	3.097×10^{-6}	0.003718	0.01702	×
single rate, $T_s = 0.2$	0.2	0.2	1	0	1.377	2.479	6.941	×
multi rate, $T_s = 0.2$	0.01	0.02	0.04835	0.04852	0.9057	0.1199	4.582	×

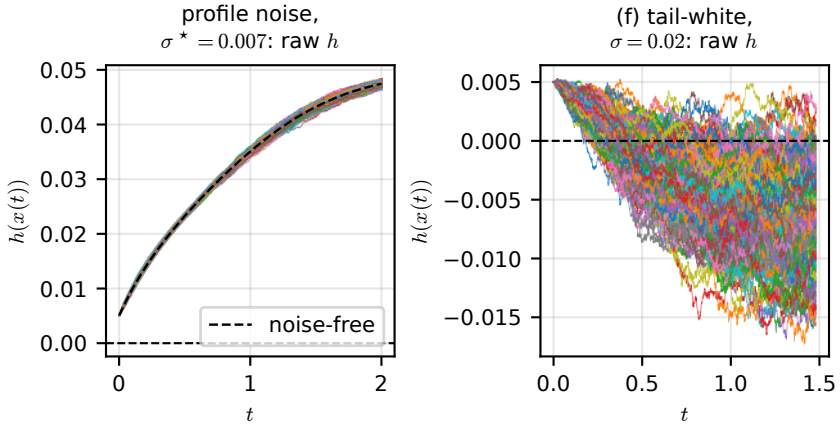


Fig. 9. Experiment 7: raw stochastic $h(x_t)$ paths for profile noise at $\sigma^\star$ with the noise-free run (left) and for ablation (f) (right).

TABLE XV
GRID SEARCH OVER THE PROFILE-NOISE INTENSITY σ (R1, $T = 2$, $\alpha = 0.05$). FLUCT. SHARE: FRACTION OF 100 SCREENING PATHS WITH $\max_t |h(x_t) - h(x_0)| \geq 0.1\, h(x_0)$; NOISE DEV.: MEDIAN OF $\max_t |h(x_t) - h_{\mathrm{det}}(t)|/h(x_0)$ (DEVIATION FROM THE NOISE-FREE RUN). $\sigma^\star = 0.007$.

σ	Corollary	failing condition	$\delta_{\mathrm{total},\sigma} + \delta_{\mathrm{stoch}} + \delta_{\mathrm{drift}}$	$\delta_{\mathrm{cert},\sigma}$	T_{TTL}	fluct. share	total dev.	noise dev.
0.002	✓	none	0.0161	0.03161	1015	1	8.399	0.04911
0.003	✓	none	0.0175	0.03161	1248	1	8.401	0.07428
0.005	✓	none	0.02084	0.03161	1667	1	8.405	0.1269
0.007	✓	none	0.02496	0.03161	2022	1	8.41	0.174
0.01	×	margin	0.03276	0.03161	2440	–	–	–
0.014	×	margin	0.04671	0.03161	2793	–	–	–
0.02	×	margin	0.07733	0.03161	2933	–	–	–

TABLE XVI
COROLLARY 1 (R1, $T = 2$, $\alpha = 0.05$, $r_{W_2} = 0$, WORST-CASE ADVERSARY, $z(0)$ ON THE LEVEL 0.9 OF $\tilde{\mathcal{C}}_r$): MONTE CARLO EXIT FREQUENCY VS. THE BOUND α WITH THE EXACT CLOPPER–PEARSON INTERVAL; CRUDE: BOUND WITH $\sigma_\perp := 0$. ON R2 THE COROLLARY REFUSES ALREADY AT $\sigma = 0$: $\gamma_0 h_{\mathrm{act}} = 0.007237$ EXCEEDS THE MARGIN $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}} = 9.994 \times 10^{-4}$.

noise	σ	$h(x_0)$	Corollary	crude	T_{TTL}	T_{TTL}/T	exits	CP 95% interval
profile, $\sigma^\star$	0.007	0.005051	✓	✓	2022	1011	0/2000	[0, 0.001843]
tail-white (f)	0.02	0.005051	×	✓	0	0	2000/2000	[0.9982, 1]

TABLE XVII
COROLLARY 1, DIFFUSION-ENLARGED DETERMINISTIC CONSTANTS (FULL VS. CRUDE BOUND); THE TAIL ENVELOPE MUST STAY BELOW $\sqrt{E_{\mathrm{safe}}} = 0.2236$.

bound	σ_r	$\sigma_\perp^2$	Δ_σ	$\varepsilon_{\mathrm{basis},\sigma}$	$\delta_{\mathrm{total},\sigma}$	$T_\perp$
profile, $\sigma^\star$, full	0.007	4.917×10^{-10}	9.703×10^{-5}	0.00739	0.002634	∞
profile, $\sigma^\star$, crude $\sigma_\perp := 0$	0.007	0	0	0.007293	0.002634	∞
tail-white (f), full	8.742×10^{-18}	0.0632	1.1	1.107	0.005408	0
tail-white (f), crude $\sigma_\perp := 0$	8.742×10^{-18}	0	0	0.007293	0.002634	∞

TABLE XVIII
COROLLARY 1, STOCHASTIC CONSTANTS OF (28) AND THE KUSHNER BOUND (FULL VS. CRUDE BOUND).

bound	r_{BM}	$\delta_{\mathrm{Itô}}$	σ_h	$\Delta\delta_{\mathrm{pw}}$	δ_{stoch}	c_{int}
profile, $\sigma^\star$, full	0.003237	4.9×10^{-5}	0.00313	0.008873	0.02074	5.146×10^{-6}
profile, $\sigma^\star$, crude $\sigma_\perp := 0$	0.003237	4.9×10^{-5}	0.00313	0.008873	0.02074	5.146×10^{-6}
tail-white (f), full	4.043×10^{-18}	8.786×10^{-35}	3.909×10^{-18}	0.001904	-0.1744	-1.837×10^{-7}
tail-white (f), crude $\sigma_\perp := 0$	4.043×10^{-18}	8.786×10^{-35}	3.909×10^{-18}	0.001896	0.009573	4.578×10^{-6}

TABLE XIX
STEP-HALVING CONVERGENCE OF THE EXPONENTIAL INTEGRATOR (OPEN-LOOP NONLINEAR LATTICE).

substeps per period	step	rel. error ETD2RK	ratio	rel. error ETD1
1	0.05	1.238×10^{-9}	–	4.081×10^{-8}
2	0.025	3.095×10^{-10}	4.002	2.048×10^{-8}
4	0.0125	7.731×10^{-11}	4.003	1.026×10^{-8}
8	0.00625	1.927×10^{-11}	4.011	5.135×10^{-9}
16	0.003125	4.765×10^{-12}	4.045	2.569×10^{-9}

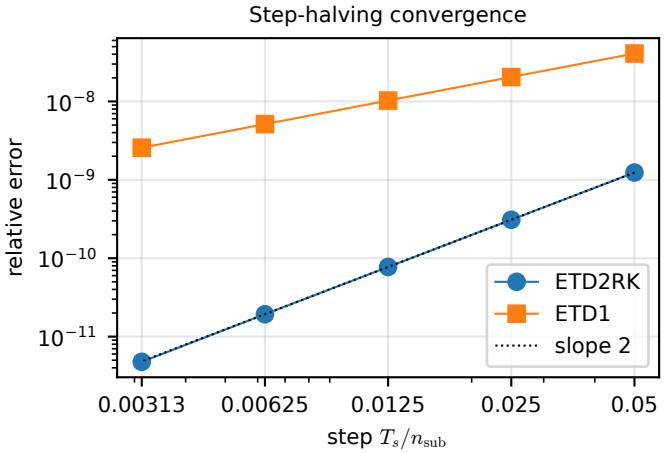


Fig. 10. Experiment 8: step-halving convergence of ETD2RK and ETD1.

TABLE XX
NUMERICAL-INTEGRITY CHECKS.

check	value
spectral radius $\max\|\lambda\|$	72
explicit Euler amplification at conv. step	3.73
stiffness ratio	43.83
latch: rel. error exact	2.808×10^{-14}
latch: rel. error grid-rounded	0.01675
modal vs nodal rel. error	1.928×10^{-14}
ETD2RK observed orders	2.001, 2.001, 2.004, 2.016
determinism SHA-256 (prefix)	`2bde49483745ec98`

TABLE XXI
MEASURED WALL-CLOCK AND PEAK RESIDENT MEMORY PER SCRIPT (2 CORES); EXPERIMENT ASSERTIONS PASSED: 248 OF 248; PYTEST: 65 PASSED.

script	wall [s]	peak RSS [MB]
exp0 openloop	18.4	214
exp1 constants	4.8	112
exp2 crown	6.9	150
exp3 delta cert	5.9	206
exp4 theorem1	52.6	343
exp4b budget	203.6	1010
exp5 deadzone	54.4	210
exp6 ablations	65.2	299
exp7 stochastic	166.4	1040
exp8 numerics	9.2	220
exp9 wsweep	31.4	194
pytest	23.7	1193
total	642.9	1193